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PROJECT REPORT

Title:- Student Performance Prediction

INTRODUCTION:-

In the field of education, predicting student performance helps in identifying weak students and improving their outcomes. This project uses Machine Learning (Linear Regression) to predict a student's final marks based on study habits and past performance.

Objectives:-

To build a model that predicts final grade (G3) of students

To analyze the effect of:

- Study time
- Absences
- Previous marks (M1, M2)

To classify students as Pass/Fail and assign grades

Dataset Description:-

- Dataset used: student-mat.csv
- Source: UCI Machine Learning Repository
- Key features used:

Feature	Description
Studytime	Weekly study time (1–8 scale)
Absences	Number of absences
G1 (M1)	First period marks
G2 (M2)	Second period marks
G3	Final marks (target variable)

Technologies Used:-

- Python
- Libraries:
- pandas → Data handling
- sklearn → Machine learning
- train_test_split
- LinearRegression

Methodology:-

Step 1: Data Loading

- Dataset is loaded using pandas
- Semicolon (;) is used as separator

Step 2: Data Preprocessing

- Selected relevant columns:studytime, absences, G1, G2, G3
- Renamed:G1 → M1

  G2 → M2

Step 3: Feature Selection

- Input features (X):studytime, absences, M1, M2
- Target variable (y):G3

Step 4: Data Splitting

- 80% → Training data
- 20% → Testing data

Step 5: Model Training

- Used Linear Regression model
- Model learns relationship between inputs and final marks

Step 6: Prediction

- User inputs:Study time,Absences,M1, M2 marks
- Model predicts final marks (G3)

Working of the Model:-

Linear Regression follows the equation:

$$G3=b0+b1(studytime)+b2(absences)+b3(M1)+b4(M2)$$

- The model calculates coefficients (b1, b2, etc.)
- Uses them to predict final marks

Output System:-

✅ Pass/Fail Criteria:

Pass: Marks ≥ 33

Fail: Marks < 33

Sample Output:-

Enter student details:

Study time: 5

Absences: 2

M1: 70

M2: 75

Predicted Final Marks: 74.5

Result: PASS

Grade: B (Very Good)

Advantages:-

- Simple and easy to implement
- Helps teachers identify weak students
- Uses real data for prediction
- Fast prediction

Limitations:-

- Linear Regression assumes linear relationship
- Accuracy depends on dataset quality
- Limited features used
- Does not consider external factors (health, environment)

Future Scope:-

- Use advanced models (Random Forest, Neural Networks)
- Add more features (internet usage, family background)
- Deploy as a web app or mobile app
- Improve accuracy with larger datasets

Challenges Faced:-

- Understanding dataset format (semicolon-separated CSV and selecting correct columns)
- Data preprocessing issues, such as renaming columns and choosing relevant features

- Model accuracy limitations due to simple Linear Regression
- Handling user input properly and ensuring correct data types
- Overfitting/underfitting concerns due to limited features
- Interpreting model output and converting it into meaningful results (grades, pass/fail)

Conclusion:-

The project successfully demonstrates how Machine Learning can be used to predict student performance. The Linear Regression model provides a simple yet effective way to estimate final marks and classify students. This system can help educators take timely actions to improve student outcomes.

What I Learned from This Project:-

- Basics of Machine Learning workflow (data → training → prediction)
- Practical use of Linear Regression algorithm
- Importance of feature selection in improving predictions
- How to use Python libraries (pandas, sklearn)
- Understanding how real-world data affects model performance
- Converting predictions into useful decision-making outputs (grades, result system)

References:-

- UCI Machine Learning Repository
- Scikit-learn Documentation
- Python Pandas Documentation